Design Document

Project: ESP32 Surge Protector
Platform: ESP32 | **Environment:** Wokwi Simulation | **Language:** Arduino C++

1. System Overview

The system monitors simulated AC voltage and load current using two potentiometers connected to the ESP32 ADC inputs. It detects overvoltage and overcurrent conditions, switches the simulated load OFF, indicates the operating state through LEDs, and attempts automatic recovery when readings become safe. Users can configure protection thresholds locally through an LCD menu or remotely through the Serial Monitor.

2. Hardware and Interface Design

Component	ESP32 Interface	Purpose
Voltage potentiometer	GPIO 34 (ADC)	Simulated voltage input, 0–300 V
Current potentiometer	GPIO 35 (ADC)	Simulated current input, 0–10 A
Simulated load output	GPIO 23	Load ON/OFF control
Normal / Fault / Recovery LEDs	GPIO 18 / 19 / 5	Operating-state indication
UP / DOWN / SET buttons	GPIO 25 / 26 / 27	Local menu navigation and configuration
16×2 I2C LCD	Address `0x27`	Display and local configuration

3. Firmware Architecture

The firmware is organized into five functional modules:

- **Sensor Acquisition:** Reads ADC values and scales them into simulated voltage and current.
- **Protection Controller:** Uses a state machine to manage normal operation, overvoltage, overcurrent, recovery waiting, and lockout.
- **Local User Interface:** Implements debounced buttons and an LCD menu for adjusting protection settings.

- **Serial Interface:** Supports status queries, configuration commands, help, and lockout reset.
- **Output and Telemetry:** Controls the simulated load and status LEDs and periodically reports readings and state.

4. Protection Logic and Configuration

Parameter	Default	Supported Range
Overvoltage threshold	250 V	20–290 V
Overcurrent threshold	5 A	0.5–9.5 A
Fault qualification	500 ms	Fixed
Recovery delay	5 s	1–60 s
Automatic recovery attempts	3	Fixed

A reading exceeding either protection threshold continuously for at least 500 ms triggers a fault and turns the simulated load OFF. Recovery begins when both readings fall below their thresholds by a hysteresis margin of 10 V and 0.5 A, respectively. After the configured stability delay, the firmware reconnects the load. After three successful automatic recovery attempts, a subsequent fault causes lockout; `RESET` clears lockout only when readings are safe.

5. Validation and Safety

The design can be tested in Wokwi by adjusting the potentiometers, changing settings through the buttons or serial commands, and observing the load output, LEDs, LCD, and telemetry. Sensor values and load switching are simulated; settings are not retained after restart. This prototype must not be connected directly to mains AC or a real appliance. A physical implementation requires isolated sensing, appropriately rated switching hardware, and independent hardware-level safety protection.